

CS3008-DEEP LEARNING

TEXT-GUIDED BRAIN TUMOR SEGMENTATION USING VISION-LANGUAGE MODELS WITH FLAIR MRI AND RADIOLOGY REPORTS

Presented by:

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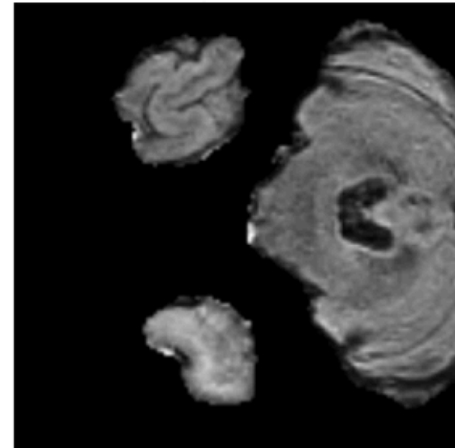
Rohit Kumar-CS23B2053

PROBLEM STATEMENT

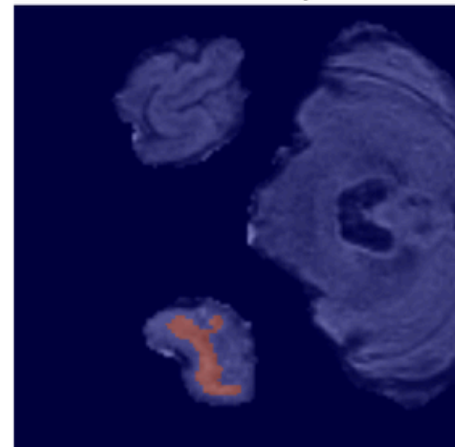
Brain tumor segmentation from Magnetic Resonance Imaging (MRI) is a critical step in clinical diagnosis, surgical planning, and post-operative monitoring. Traditional deep learning approaches rely entirely on visual feature extraction. However, in clinical practice, radiologists evaluate MRI scans in conjunction with prior knowledge, physiological hypotheses, and written clinical descriptions (e.g., "ring-enhancing lesion with surrounding edema in the left temporal lobe").

Dataset Samples: 5-Slice Center FLAIR & Tumor Mask

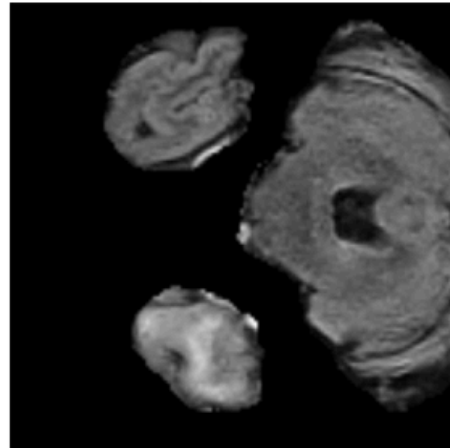
Sample 1 FLAIR



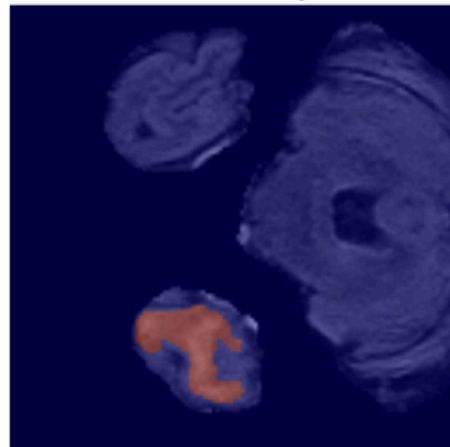
Tumor Overlay



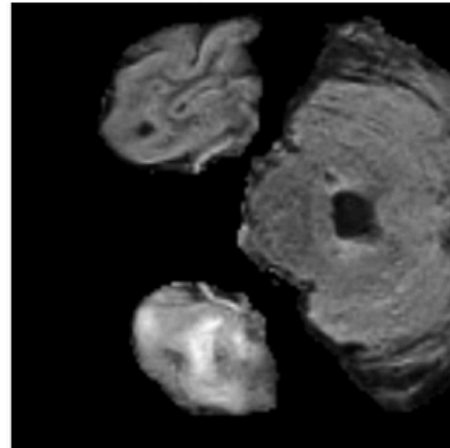
Sample 2 FLAIR



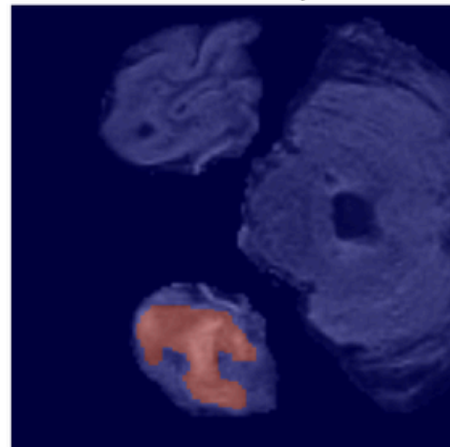
Tumor Overlay



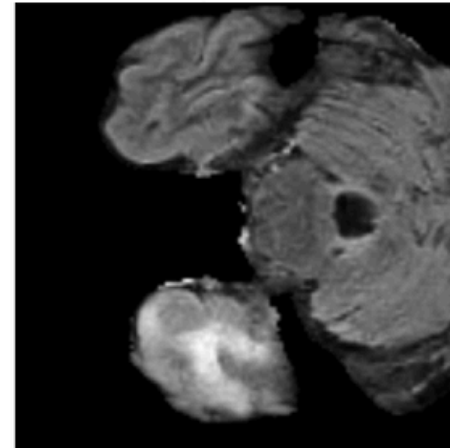
Sample 3 FLAIR



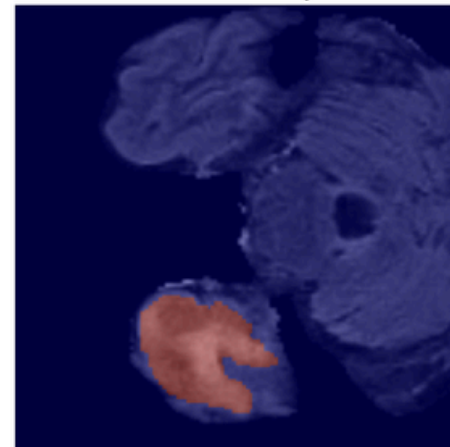
Tumor Overlay



Sample 4 FLAIR



Tumor Overlay



4

Demonstrate via strict ablation studies that Natural Language processing explicitly improves segmentation metrics (Dice, IoU, Hausdorff) over a perfectly identical imageonly baseline.

1

Take multi-slice FLAIR MRI volumes and their corresponding patient-level radiology text descriptions as joint input.

OBJECTIVES

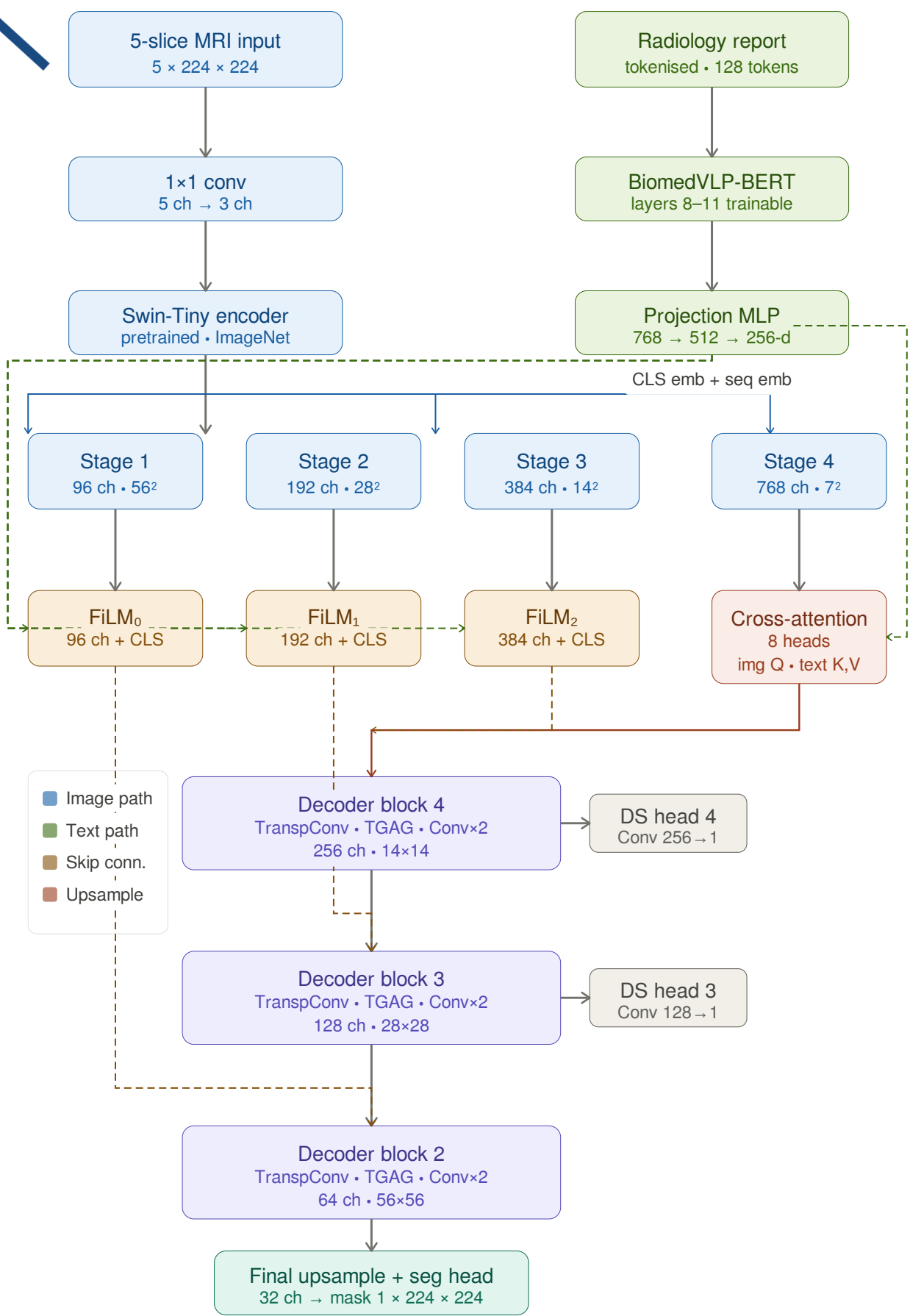
3

Perform pixel-wise binary tumor segmentation via a text-conditioned decoder utilizing Deep Supervision.

2

Learn a shared image-text latent representation using cross-attention and FiLM fusion.

ARCHITECTURE OVERVIEW



The multimodal architecture, termed TG-SwinUNet (Text-Guided Swin-UNet), represents a highly sophisticated Vision-Language fusion network designed to explicitly condition spatial segmentation on clinical language.

Macro Architecture: Overview flow diagram of the TG-SwinUNet multimodal architecture. The visual pathway processes 5-channel MRI slices using a Swin Transformer, while the text pathway processes clinical reports. Textual embeddings modulate the visual features via FiLM at shallow layers and via Cross-Attention at the bottleneck.

ARCHITECTURE BREAKDOWN

1. Image Encoder (Swin Transformer):

A swin_tiny_patch4_window7 backbone. The 5-slice input is projected to 3 channels via a 1×1 convolution. The encoder extracts hierarchical spatial features at four resolutions: 562, 282, 142, and 72.

2. Text Encoder (BioViL-T): We utilise microsoft/BiomedVLP-CXR-BERT-specialized. To prevent catastrophic forgetting while allowing domain adaptation, layers 0–7 are strictly frozen, while layers 8–11 are unfrozen and trained at a lower learning rate. The model outputs both a global CLS embedding and a full sequence of 128 token embeddings.

Four Subsystems

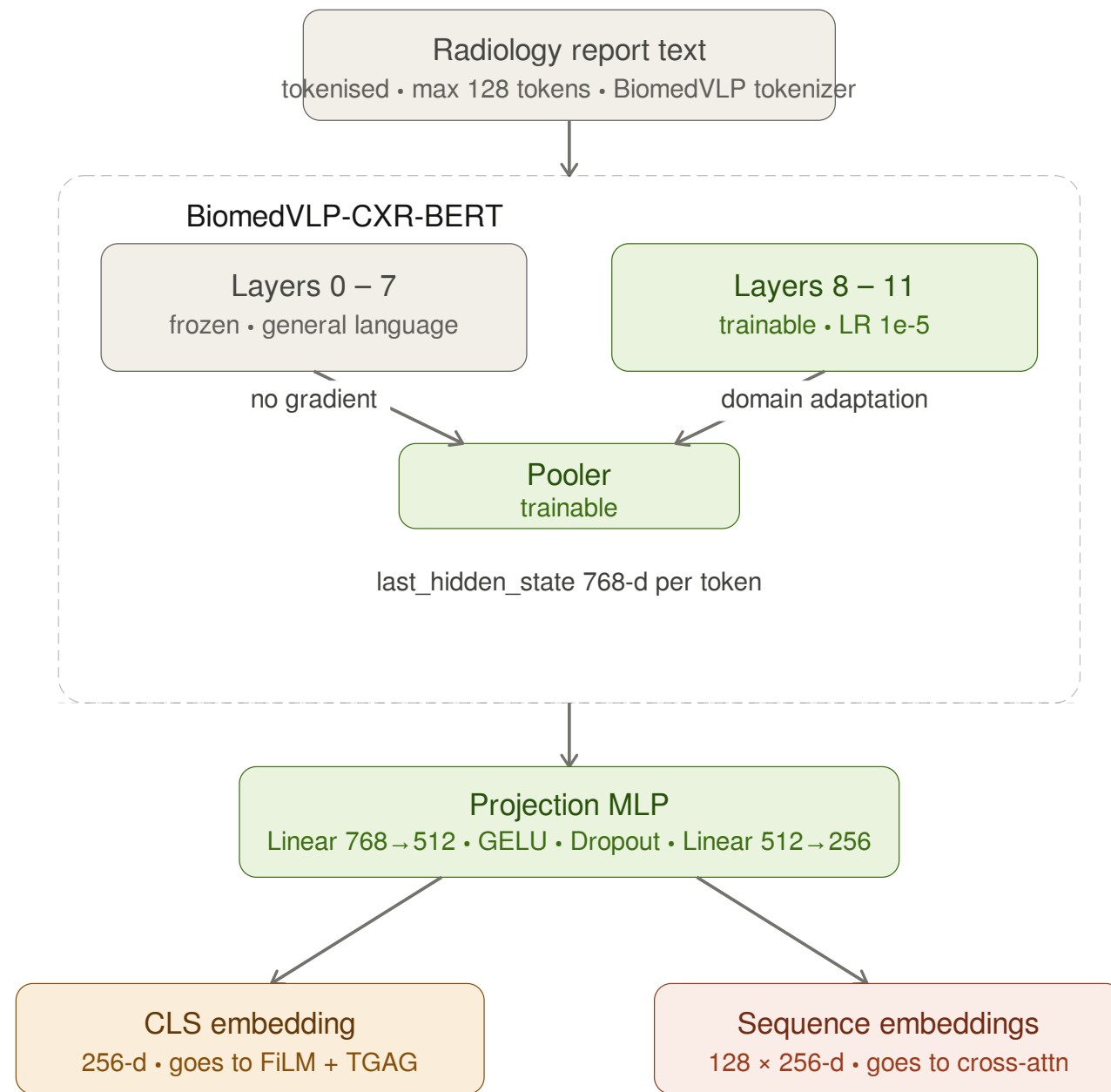
3. Fusion Mechanisms:

- **Multi-Level FiLM:** Feature-wise Linear Modulation applies scaling (γ) and shifting (β) to the image features at the first three skip connections using the global CLS token. Mathematically: $\text{FiLM}(F, \text{ecls}) = \gamma(\text{ecls}) \odot F + \beta(\text{ecls})$.
- **Cross-Attention Bottleneck:** At the deepest 7×7 bottleneck, spatial image features act as Queries, while the 128 text tokens act as Keys and Values. This allows explicit spatial regions to attend to specific clinical words.

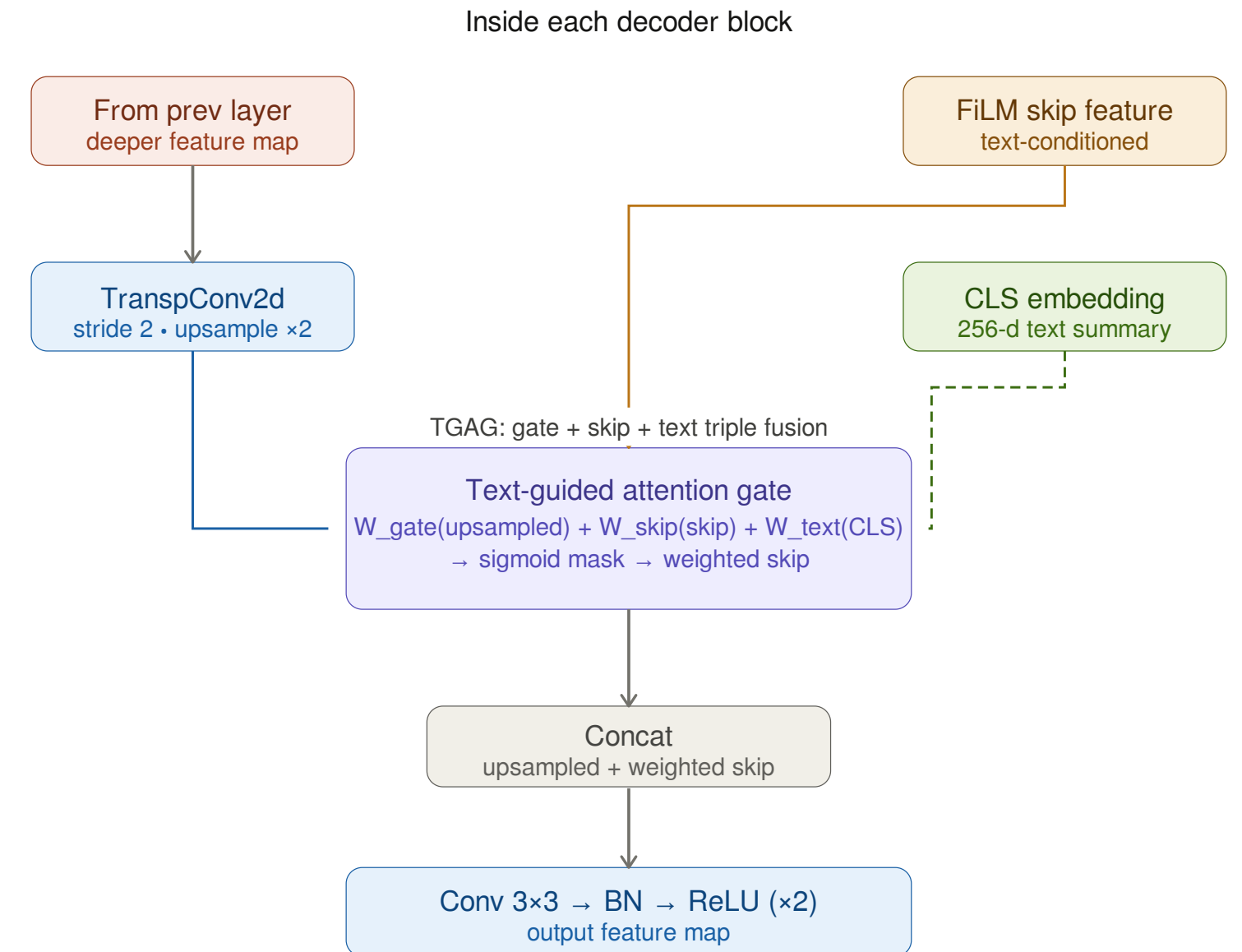
4. Decoder with TGAG & Deep Supervision: A standard U-Net transposed convolution decoder is fundamentally upgraded with two mechanisms:

- **Text-Guided Attention Gates (TGAG):** TGAG utilizes the text 'CLS' token to explicitly condition which visual features to suppress or amplify during skipconnection upsampling.
- **Deep Supervision:** Auxiliary segmentation heads are attached to intermediate decoder blocks (dec3 and dec4). This forces early layers to learn semantic tumor boundaries directly.

ARCHITECTURE BREAKDOWN

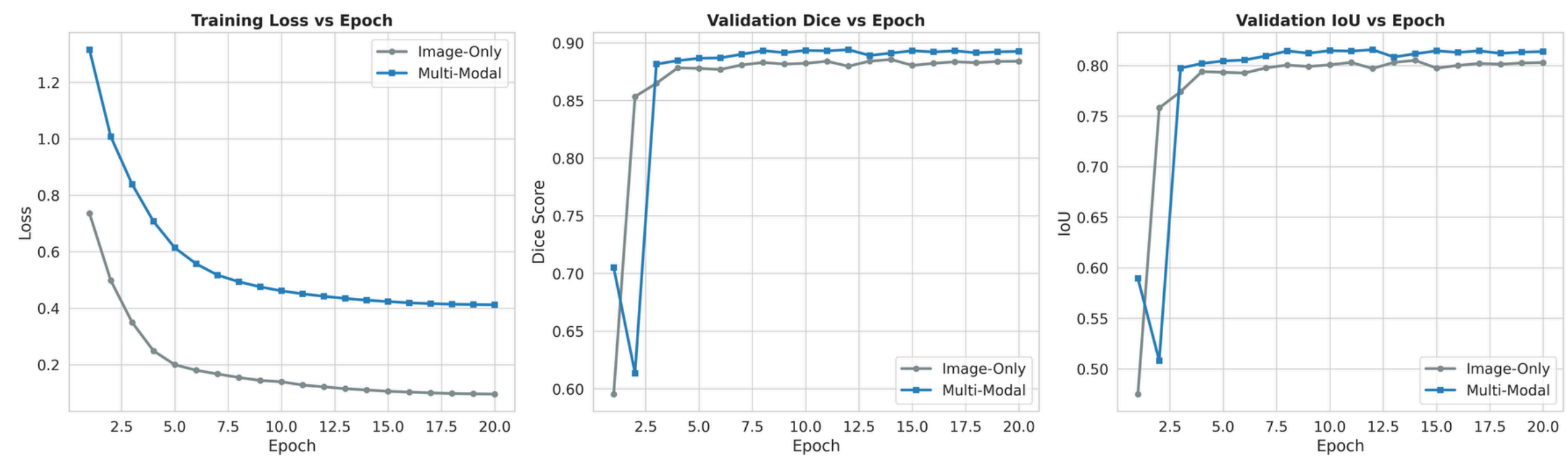
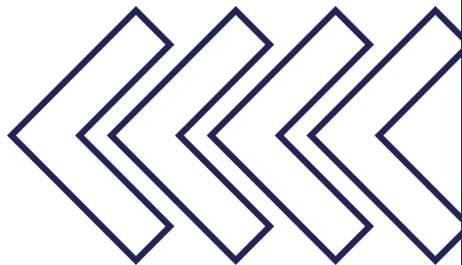


Text Pathway: Detailed breakdown of the BioViL-T text encoder, highlighting the partial unfreezing strategy for domain adaptation and the Projection MLP that maps the 768-D embeddings down to 256-D for fusion.



Micro Architecture: Detailed breakdown of the Text-Guided Attention Gate (TGAG) inside the decoder. This demonstrates mathematically how the textual CLS embedding is fused with the upsampled and skip features to generate a conditioning mask.

Both the Image-Only baseline and the Multimodal model were trained from scratch for 20 epochs to ensure a perfectly fair ablation study



Training metrics across 20 epochs. The multimodal architecture consistently demonstrates superior validation Dice and IoU scores compared to the image-only baseline.

Hyperparameter	Image-Only	TG-SwinUNet (MM)
Batch size	16	16
Epochs	20	20
Optimiser	AdamW	AdamW (dual LR)
LR (CNN Base)	2×10^{-4}	1×10^{-4}
LR (BERT layers 8–11)	—	1×10^{-5}
LR scheduler	Cosine Annealing	Cosine Annealing
Mixed precision	AMP (fp16)	AMP (fp16)

TRAINING OVERVIEW



The total loss is a weighted combination of four terms explicitly designed for this task:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Seg}}(\text{Final}) + 0.5\mathcal{L}_{\text{Seg}}(\text{Dec3}) + 0.25\mathcal{L}_{\text{Seg}}(\text{Dec4}) + 0.2\mathcal{L}_{\text{Boundary}} + 0.1\mathcal{L}_{\text{Contrastive}}$$

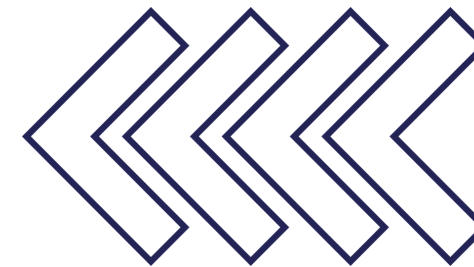
where LSeg is the sum of Dice Loss and Focal Loss.

Dice Loss optimises the volume overlap:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2|P \cap T| + 1}{|P| + |T| + 1}$$

Focal Loss penalises hard, class-imbalanced pixels:

$$\mathcal{L}_{\text{Focal}} = -\alpha(1 - p)^\gamma \log p$$



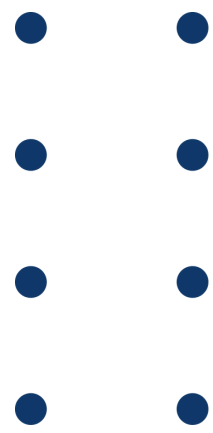
Boundary Loss: A morphological boundary map B is extracted via Max-Pooling. Prediction errors at the edge receive a 6× penalty, optimizing Hausdorff Distance.

Contrastive Loss (CLIP-style): A temperature-scaled cosine similarity loss enforces alignment between the global image embedding and the text embedding, pulling matching modalities into a shared latent space.

LOSS FUNCTIONS



EVALUATION RESULTS AND METRICS

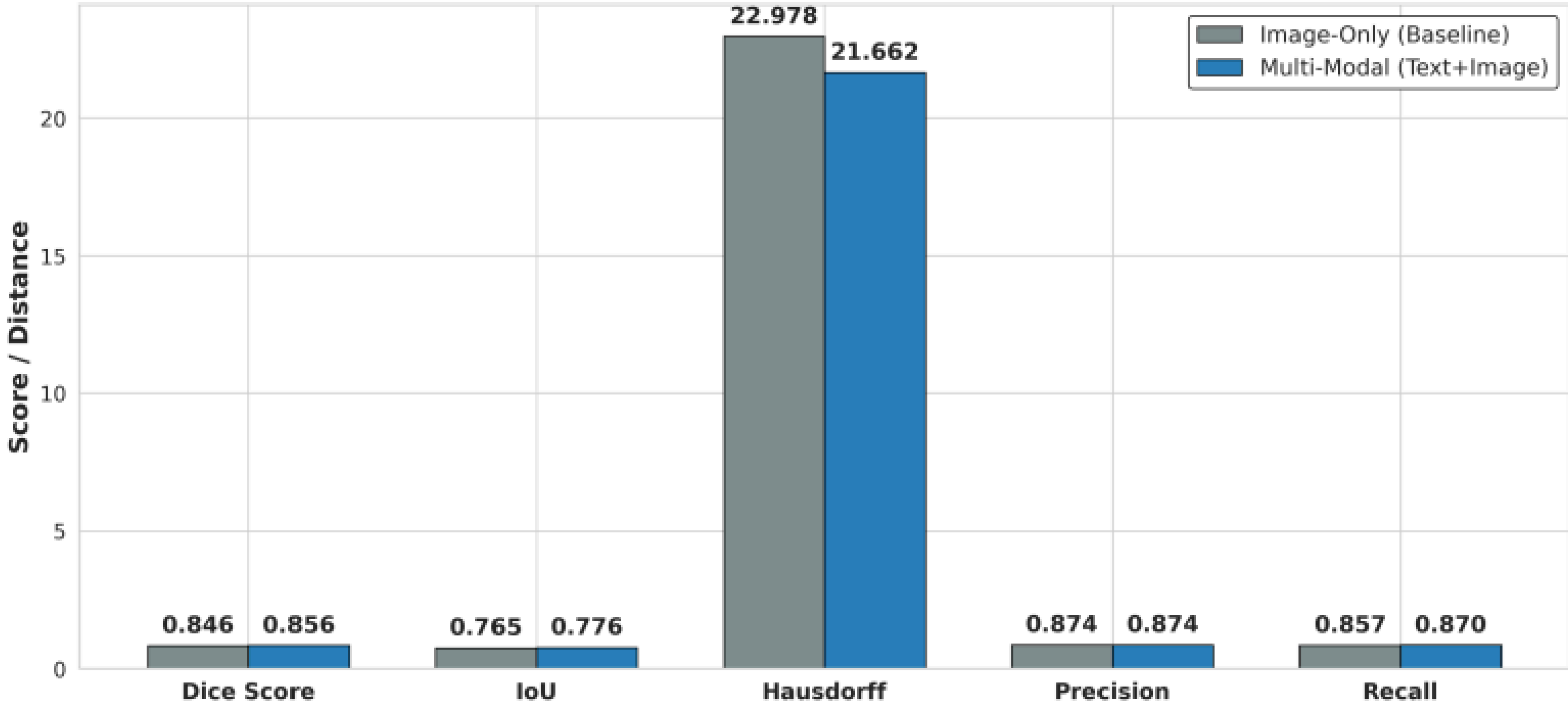


Evaluation Metrics

Model	Dice Score	IoU	Hausdorff	Precision	Recall
Image-Only Baseline	0.846	0.765	22.978	0.874	0.857
Multimodal (TG-SwinUNet)	0.856	0.776	21.662	0.874	0.870
Improvement	+0.010	+0.011	-1.316	0.000	+0.013

The final evaluation was conducted on the complete validation set.

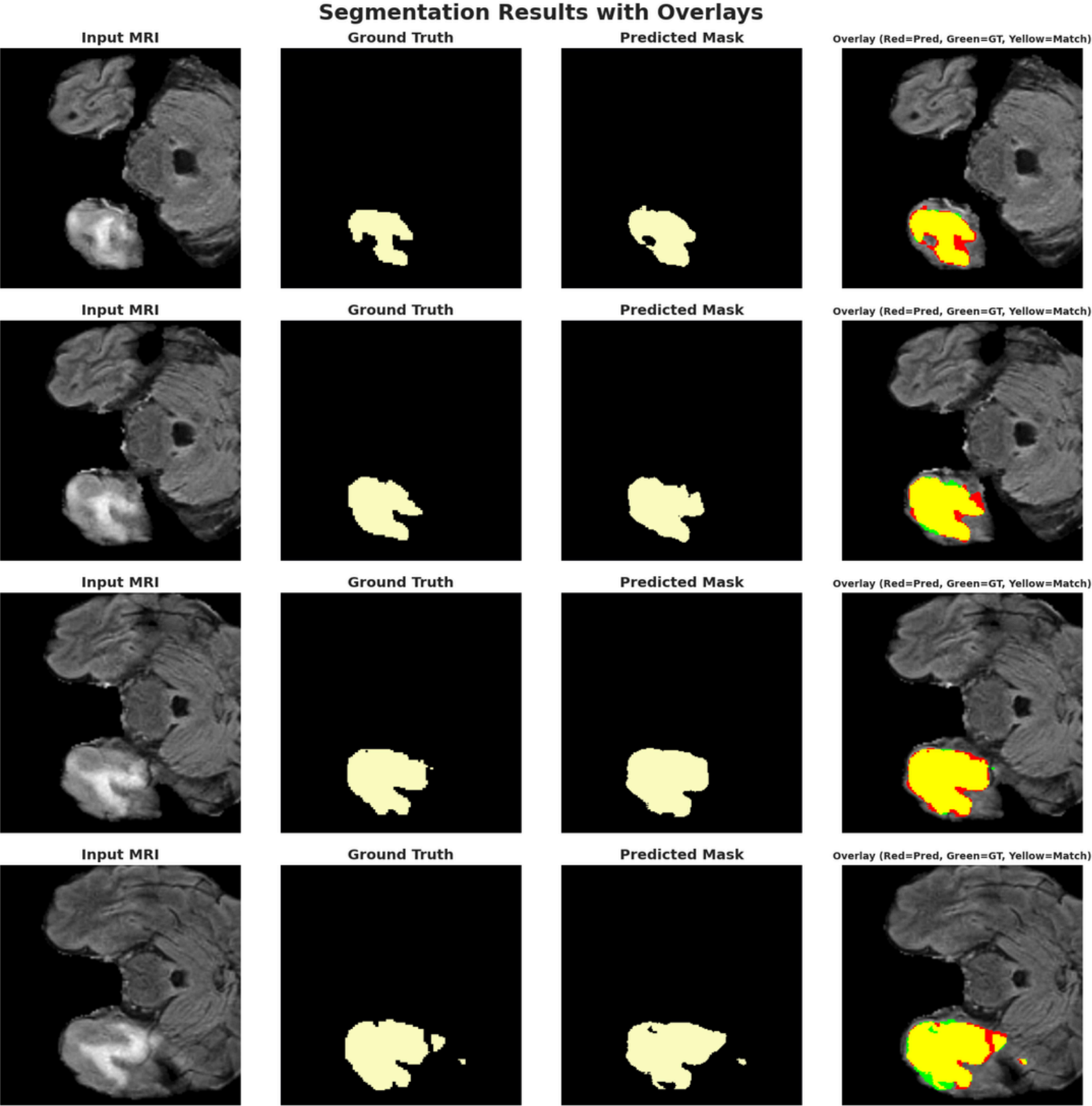
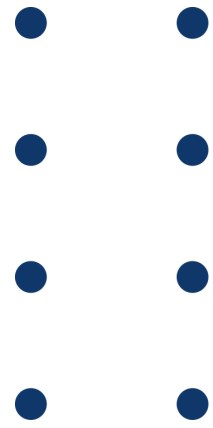
Ablation Study: Performance Comparison



Ablation study bar chart demonstrating the distinct performance gap between the image-only and multimodal frameworks across all evaluated metrics.



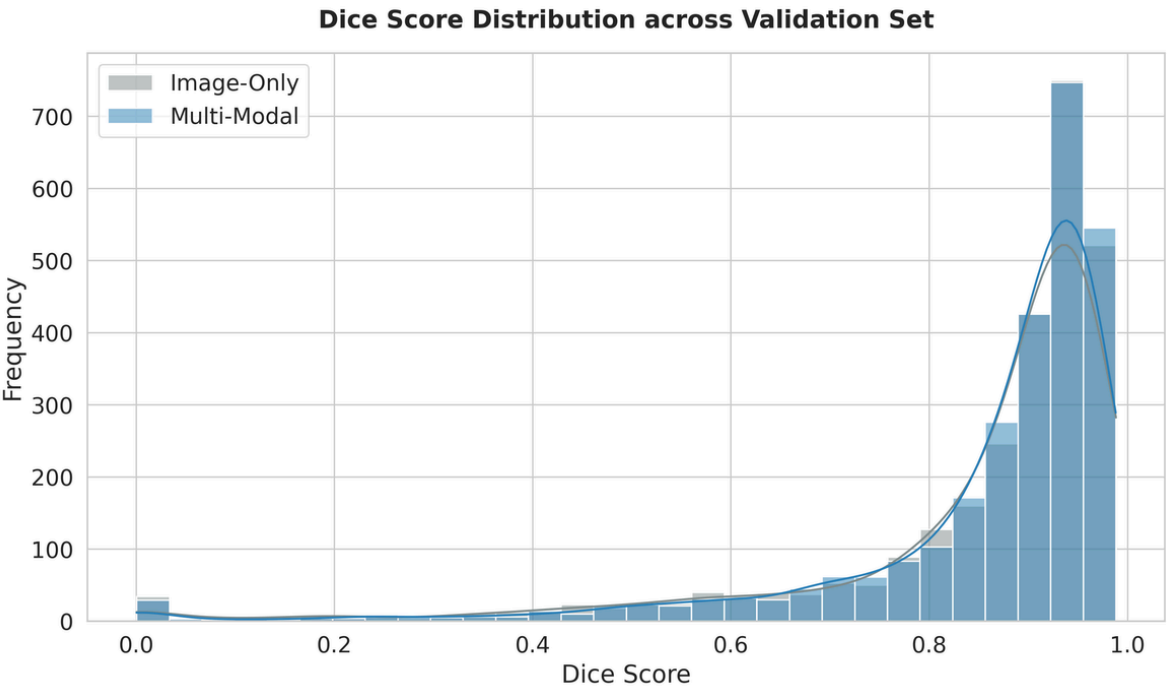
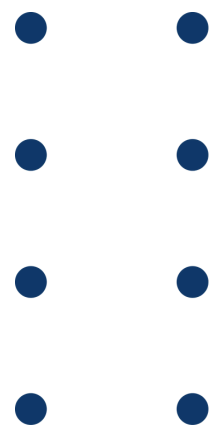
SEGMENTATION OVERLAYS AND VISUAL PREDICTIONS



Segmentation Results:
Column 1: Input MRI.
Column 2: Ground Truth.
Column 3: Predicted Mask.
Column 4: RGB Overlay Visualization (Green = Ground Truth, Red = Prediction, Yellow = Intersection).

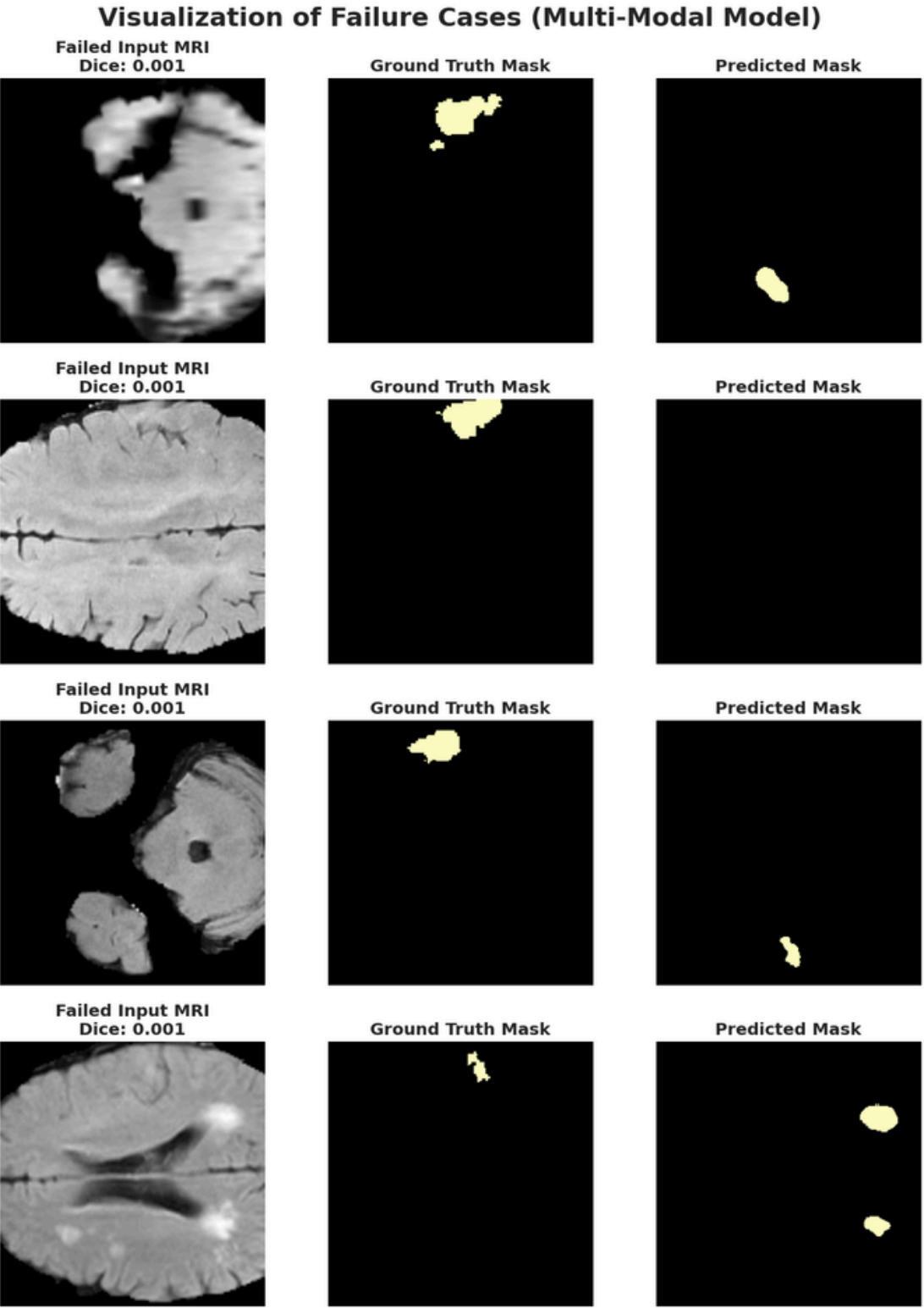


ERROR ANALYSIS AND DISTRIBUTIONS

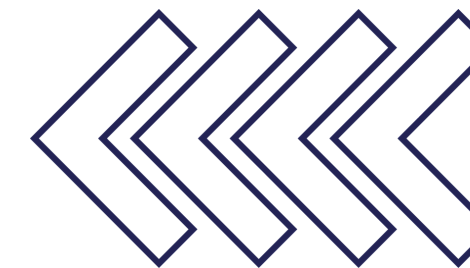
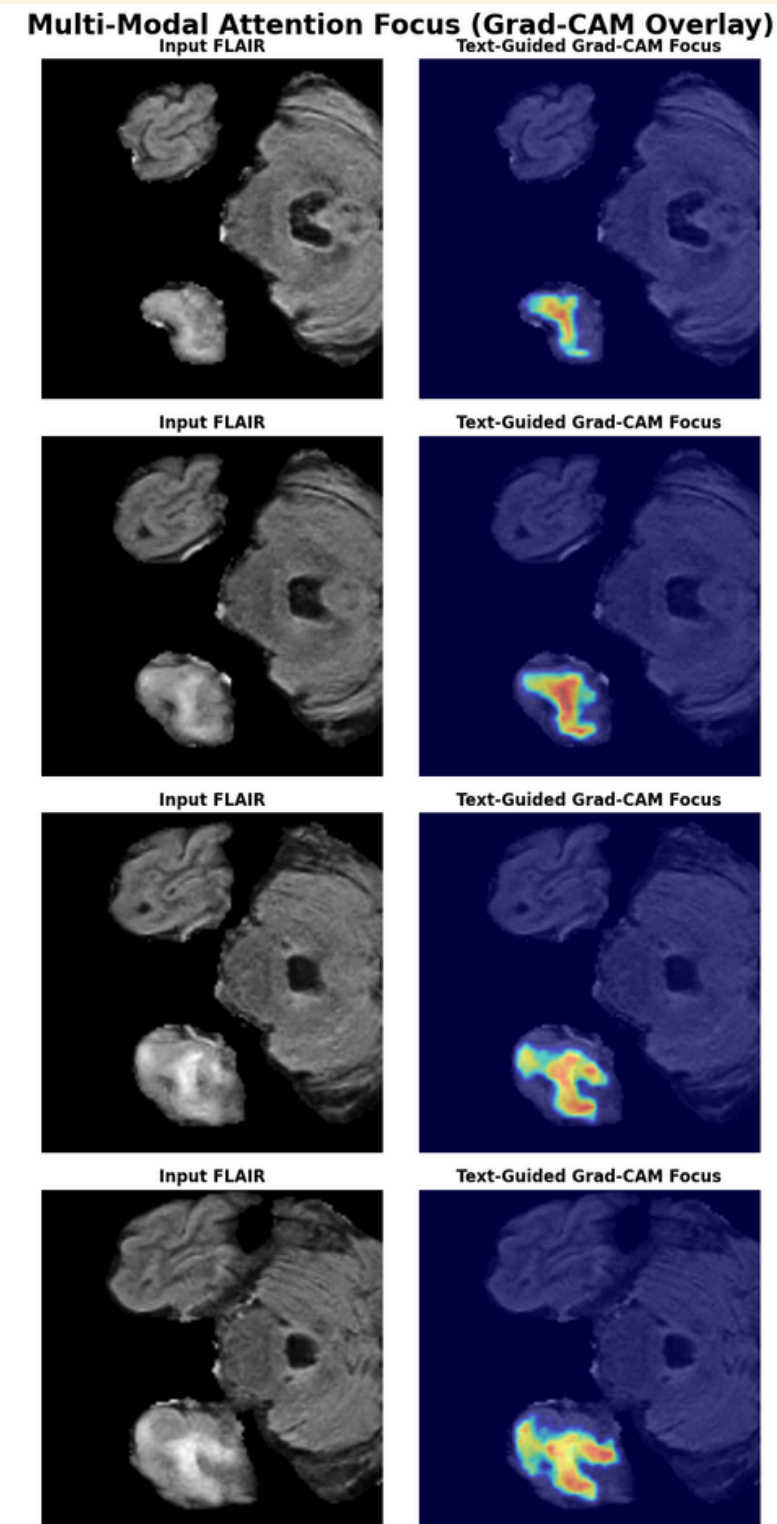


Left: KDE Density plot showing the validation set Dice score distribution.

Right: Visualization of Failure Cases (predictions where Dice < 0.60), allowing for explicit morphological error analysis



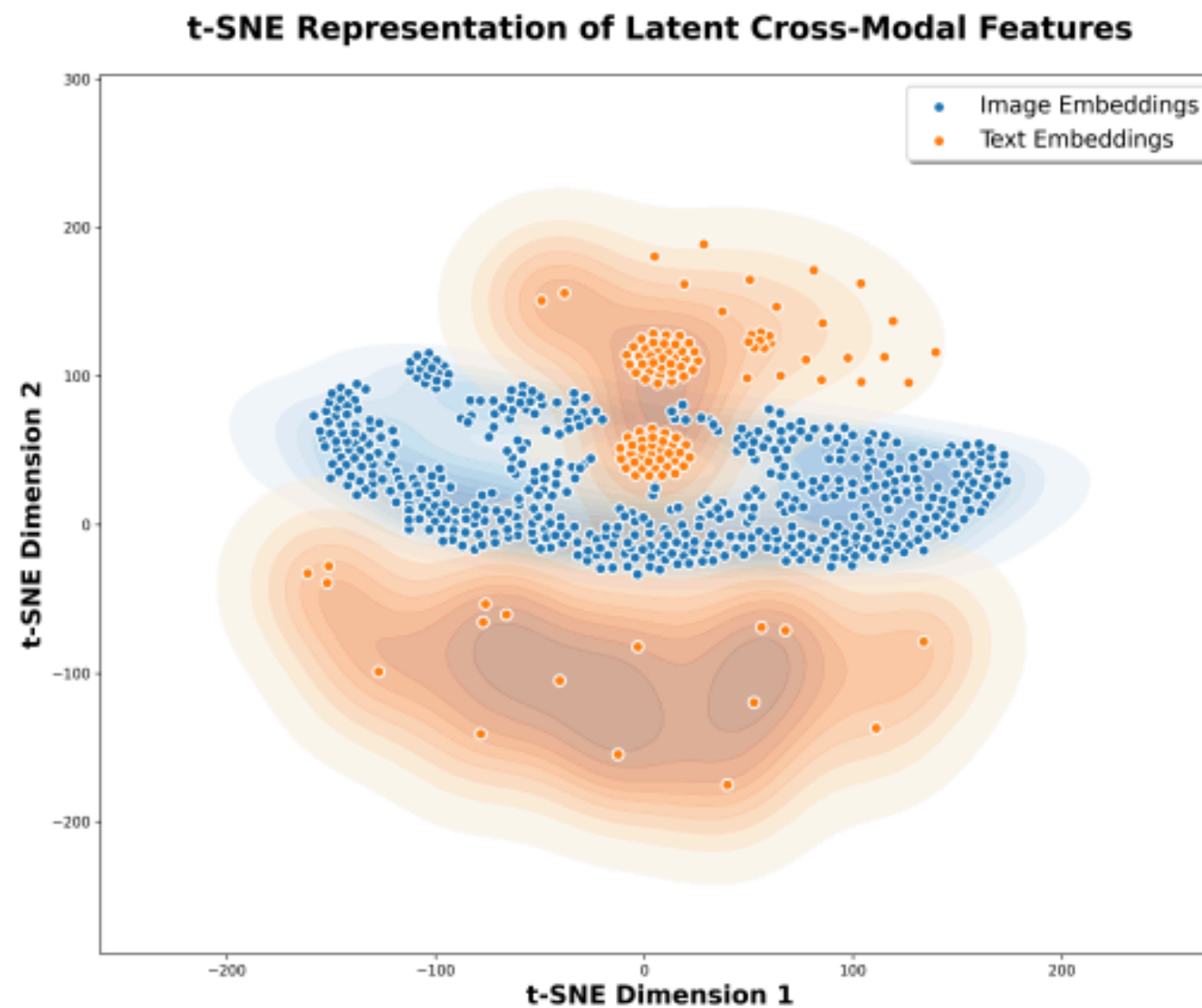
To verify that the model is actively utilising the textual clinical reports, Gradientweighted Class Activation Mapping (Grad-CAM) was applied to the high-resolution dec2 upsampling layer.



GRADCAM EXPLAINABILITY



Grad-CAM analysis. The true-color JET heatmaps demonstrate the multimodal model's focal attention on distinct morphological regions as guided by the textual embeddings.



T-SNE VISUALISATION

t-SNE clustering of Visual and Semantic embeddings. The distinct overlap and proximity of the two modalities physically proves the success of the CLIP-style Contrastive Alignment Loss. The resulting projection clearly demonstrates that the Text features and Image features converge towards each other, creating proximate clusters. Without the Contrastive Alignment Loss ($L_{Contrastive}$), these modalities would remain entirely disparate in vector space.



THANK YOU!

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